SDA Quiz 1 - 4A

14/ Nice.

Roll No: Name:

Q1. The following two code snippets count the number of people aged below 30 from the entire population of Pakistan. You want to parallelise so that multiple threads/processes can participate in the counting.

Code Snippet A:	Code Snippet B:

int count = 0;
for (Person p : population) {
if (p.getAge() < 30) {
count++;
}
}
	long count = population.stream()
	.filter(p -> p.getAge() < 30)
	.count();

Which code snippet is harder to parallelise? A 5/

Why? the 'if' with 'for' makes it harder. It is said to be the root of problems. The main reason is that there is a single variable 'count' which is being incremented in iterations. Changing its value in parallel can cause issues. So, is hard to parllelise.

Q2. The following C++ code represents a Scanner that is hard-coded to output only to PDF. This design is rigid; if you want to support saving to PNG, you would have to modify the Scanner class itself.

```
class PDFWriter {
public:
    void write(std::string data) { /* Logic to create a PDF file */ }
};

class Scanner {
    PDFWriter writer;
public:
    void scanDocument() {
        std::string rawData = "Raw Image Data";
        writer.write(rawData);
    }
};
```

Rewrite the code to make the Scanner not depend on any class that outputs in a specific format. Assume that the Scanner class will get the writer object in its constructor. Also add another class that writes in PNG format.

```
class Writer { //parent class
public:
    ~~writer();~~
virtual void write(string data) = 0;
};

class PNGWriter : public Writer {
public:
  void write(string data) override
  { //logic to save PNG; }
};
/*
Similarly, the PDF writer class
is child of the parent
class writer, and has its
function overridden */

class Scanner {
  Writer* writer;
public:
  Scanner(Writer* obj) {
    writer = obj; }
  void Scandoc() {
  String data: "Raw data";
  //we can take it as an
  //input instead of hardquote
  // but wasn't asked to do so.
  writer -> write(data);
  } //function brace
};
```

Now, through Polymorphism, we have an ~~abs~~ abstract parent class "writer" which interacts with "scanner".

We don't need to change the scanner class and can easily add more writers e.g. for jpegs etc.

Scanner class is independant now.

PDF ?

if its not clear, I wrote this :-

```
void Scandoc() {
```